

TITLE:

Prescribing trends analysis

Dataset Used:

psych-ther-ann-rep-csv-2024-25-psych-meds-2

SUBMITTED BY:

VALLURU SANDEEP KUMAR

DATA ANALYTICS INTERN

SUBMITTED TO:

ZENO TALENT

Dataset Introduction

This dataset contains **psychotropic medication prescribing status data** reported across NHS organizations such as:

- **Commissioning Regions**
- **Integrated Care Boards (ICB/CCG equivalent)**
- **Providers**
- **England overall**

The dataset includes medication prescribing information for psychological therapy patients, showing whether patients were:

- **Prescribed and taking medication**
- **Prescribed but not taking medication**
- **Not prescribed medication**
- **Unknown / invalid status**

This analysis helps identify **prescribing trends in antidepressant/anxiolytic medications by CCG (ICB)** and compare prescribing patterns across regions.

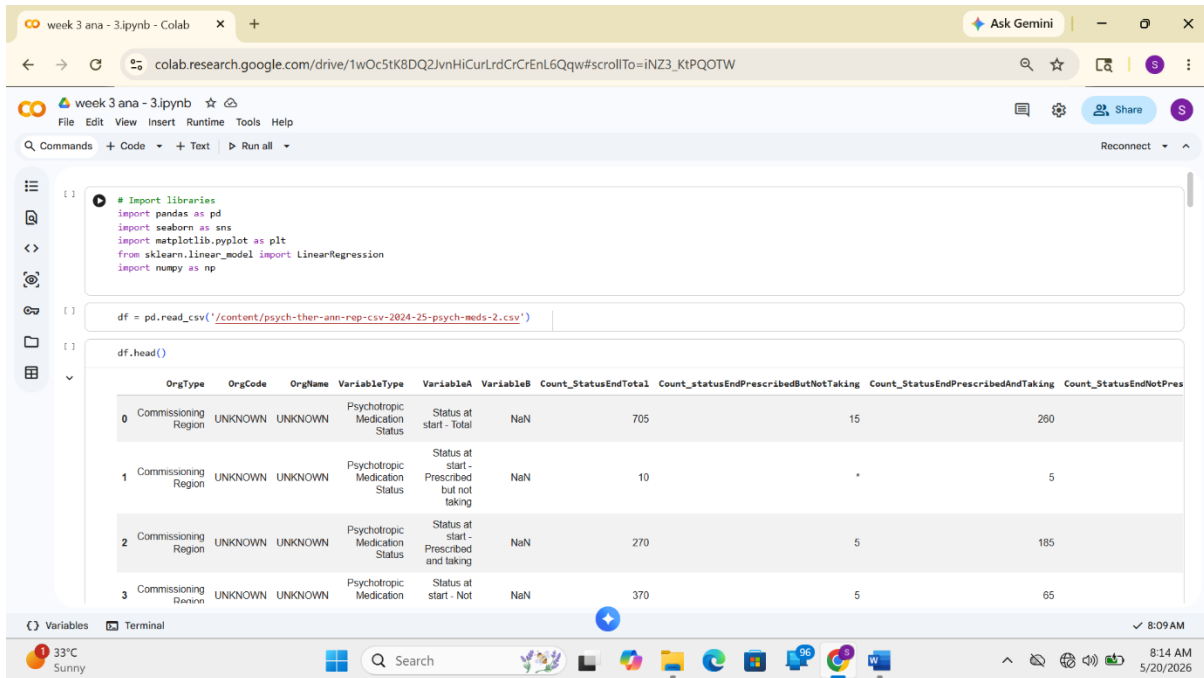
Key columns in the dataset:

- `OrgType` → NHS organization type (ICB, Provider, England, Region)
- `OrgCode` → organization code
- `OrgName` → organization name
- `VariableA` → medication status category
- `Count_StatusEndTotal` → total patients
- `Count_StatusEndPrescribedAndTaking` → prescribed and taking medication
- `Count_statusEndPrescribedButNotTaking` → prescribed but not taking
- `Count_StatusEndNotPrescribed` → not prescribed
- `Count_StatusEndNotStatedNotKnownInvalid` → unknown/invalid

Technologies Used

- **Python** → data analysis programming language
- **Pandas** → data cleaning, aggregation, trend analysis
- **Seaborn** → statistical visualizations and trend charts
- **Scikit-learn** → trend modelling / regression analysis
- **Google Colab Notebook** → cloud-based notebook environment for running Python code

Prescribing trends analysis



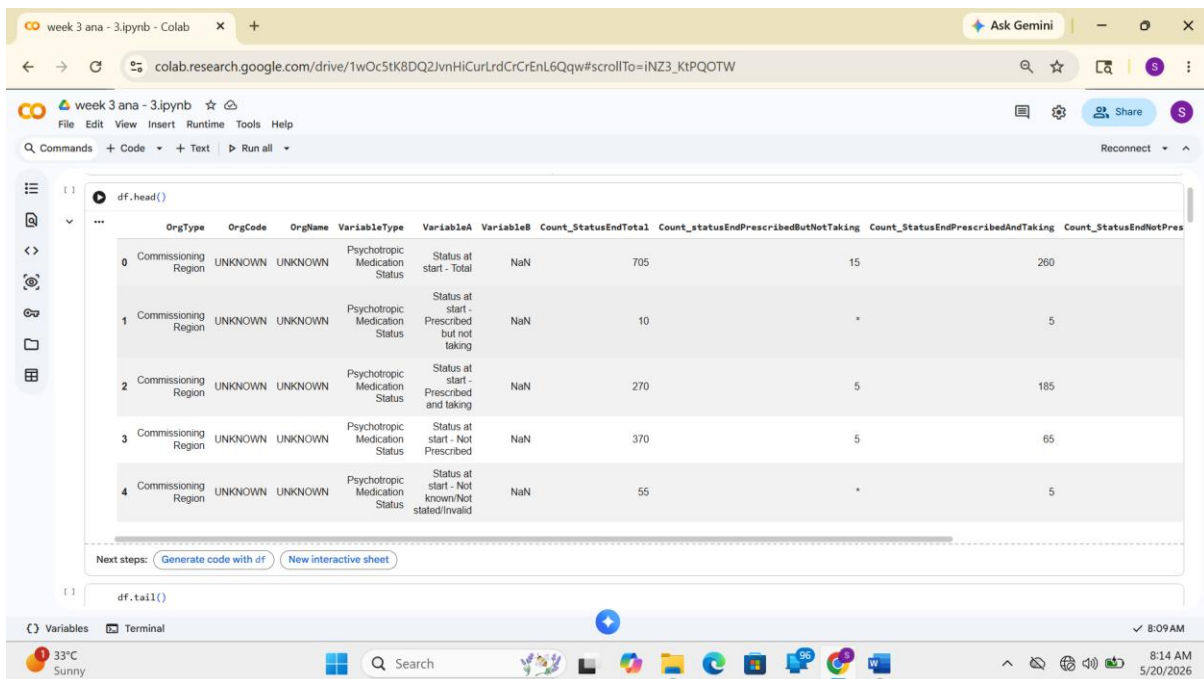
This screenshot shows a Google Colab notebook with the following code and output:

```
# Import libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
import numpy as np

df = pd.read_csv('/content/psych-ther-ann-rsp-2024-25-psych-meds-2.csv')

df.head()
```

	OrgType	OrgCode	OrgName	VariableType	VariableA	VariableB	Count_StatusEndTotal	Count_statusEndPrescribedButNotTaking	Count_StatusEndPrescribedAndTaking	Count_StatusEndNotPres
0	Commissioning Region	UNKNOWN	UNKNOWN	Psychotropic Medication Status	Status at start - Total	NaN	705	15	260	
1	Commissioning Region	UNKNOWN	UNKNOWN	Psychotropic Medication Status	Status at start - Prescribed but not taking	NaN	10	*	5	
2	Commissioning Region	UNKNOWN	UNKNOWN	Psychotropic Medication Status	Status at start - Prescribed and taking	NaN	270	5	185	
3	Commissioning Region	UNKNOWN	UNKNOWN	Psychotropic Medication Status	Status at start - Not Prescribed	NaN	370	5	65	



This screenshot shows the same Colab notebook with the following code and output:

```
df.head()
```

	OrgType	OrgCode	OrgName	VariableType	VariableA	VariableB	Count_StatusEndTotal	Count_statusEndPrescribedButNotTaking	Count_StatusEndPrescribedAndTaking	Count_StatusEndNotPres
0	Commissioning Region	UNKNOWN	UNKNOWN	Psychotropic Medication Status	Status at start - Total	NaN	705	15	260	
1	Commissioning Region	UNKNOWN	UNKNOWN	Psychotropic Medication Status	Status at start - Prescribed but not taking	NaN	10	*	5	
2	Commissioning Region	UNKNOWN	UNKNOWN	Psychotropic Medication Status	Status at start - Prescribed and taking	NaN	270	5	185	
3	Commissioning Region	UNKNOWN	UNKNOWN	Psychotropic Medication Status	Status at start - Not Prescribed	NaN	370	5	65	
4	Commissioning Region	UNKNOWN	UNKNOWN	Psychotropic Medication Status	Status at start - Not known/Not stated/Invalid	NaN	55	*	5	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.tail()
```

Colab interface showing a Jupyter Notebook with the following code and output:

```
df.tail()
```

	OrgType	OrgCode	OrgName	VariableType	VariableA	VariableB	Count_StatusEndTotal	Count_StatusEndPrescribedButNotTaking	Count_StatusEndPrescribedAndTaking	Count_StatusEndNotPrescribed
990	Provider	ZZE3K	SUNDERLAND MIND	Psychotropic Medication Status	Status at start - Total	NaN	570	*		345
991	Provider	ZZE3K	SUNDERLAND MIND	Psychotropic Medication Status	Status at start - Prescribed but not taking	NaN	10	*		*
992	Provider	ZZE3K	SUNDERLAND MIND	Psychotropic Medication Status	Status at start - Prescribed and taking	NaN	315	*		265
993	Provider	ZZE3K	SUNDERLAND MIND	Psychotropic Medication Status	Status at start - Not Prescribed	NaN	245	*		75
994	Provider	ZZE3K	SUNDERLAND MIND	Psychotropic Medication Status	Status at start - Not known/Not stated/Invalid	NaN	5	*		*

```
df.shape
```

```
(995, 11)
```

Variables: Terminal

8:09 AM

Colab interface showing a Jupyter Notebook with the following code and output:

```
df.shape
```

```
(995, 11)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 995 entries, 0 to 994
Data columns (total 11 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   OrgType                              995 non-null    object
 1   OrgCode                              995 non-null    object
 2   OrgName                              995 non-null    object
 3   VariableType                         995 non-null    object
 4   VariableA                            995 non-null    object
 5   VariableB                            0 non-null      float64
 6   Count_StatusEndTotal                 995 non-null    object
 7   Count_StatusEndPrescribedButNotTaking 995 non-null    object
 8   Count_StatusEndPrescribedAndTaking    995 non-null    object
 9   Count_StatusEndNotPrescribed          995 non-null    object
10   Count_StatusEndNotStatedNotKnownInvalid 995 non-null    object
dtypes: float64(1), object(10)
memory usage: 85.6+ KB
```

```
# Replace suppressed values (*) with NaN
df.replace('*', np.nan, inplace=True)
```

Variables: Terminal

8:09 AM

Colab interface showing code for data processing:

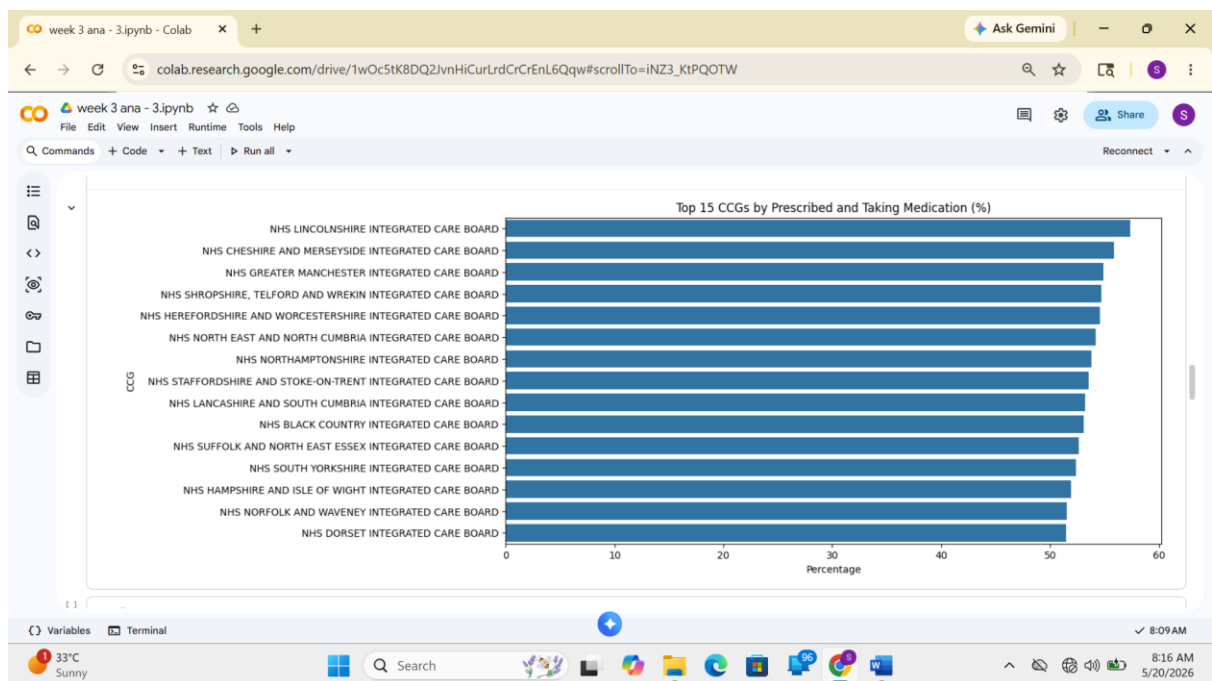
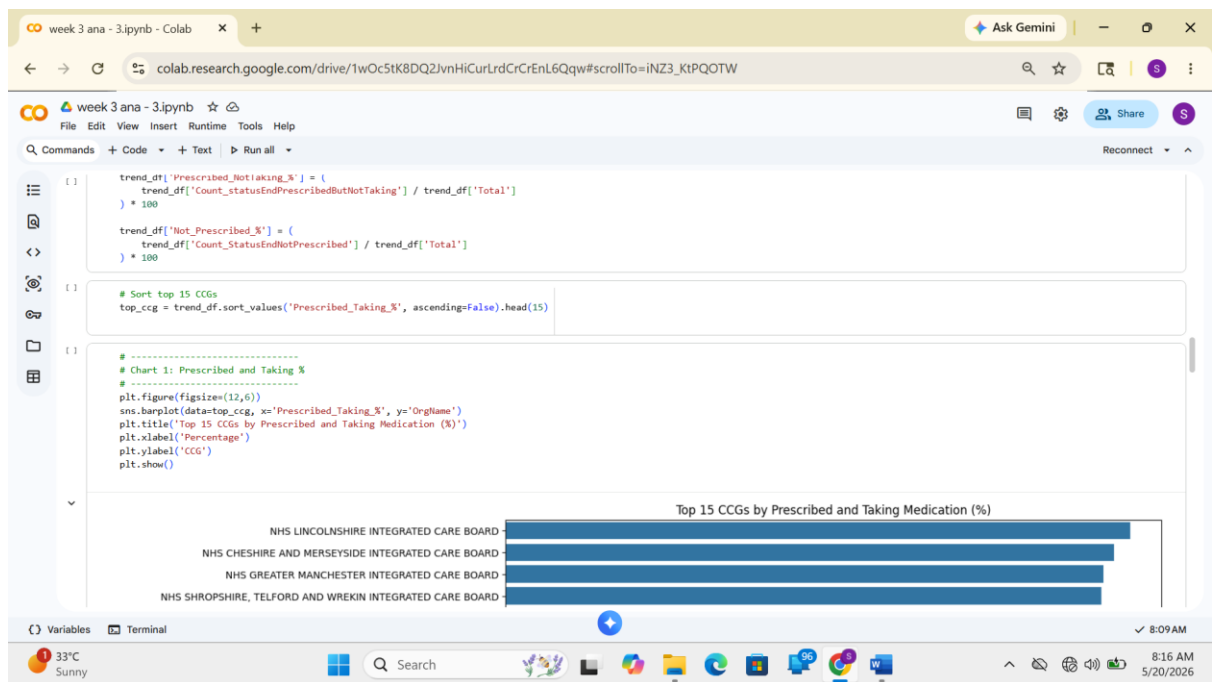
```
1 # Convert numeric columns
2 num_cols = [
3     'Count_StatusEndTotal',
4     'Count_StatusEndPrescribedButNotTaking',
5     'Count_StatusEndPrescribedAndTaking',
6     'Count_StatusEndNotPrescribed',
7     'Count_StatusEndNotStatedNotKnownInvalid'
8 ]
9
10 for col in num_cols:
11     df[col] = pd.to_numeric(df[col], errors='coerce')
12
13 # Filter only ICB (CCG equivalent)
14 ccg_df = df[df['OrgType'] == 'ICB']
15
16 # Aggregate prescribing counts by CCG
17 trend_df = ccg_df.groupby('OrgName')[[
18     'Count_StatusEndPrescribedAndTaking',
19     'Count_StatusEndPrescribedButNotTaking',
20     'Count_StatusEndNotPrescribed'
21 ]].sum().reset_index()
22
23 # Total patients for prescribing rate
24 trend_df['Total'] = (
25     trend_df['Count_StatusEndPrescribedAndTaking'] +
26     trend_df['Count_StatusEndPrescribedButNotTaking'] +
27     trend_df['Count_StatusEndNotPrescribed']
28 )
```

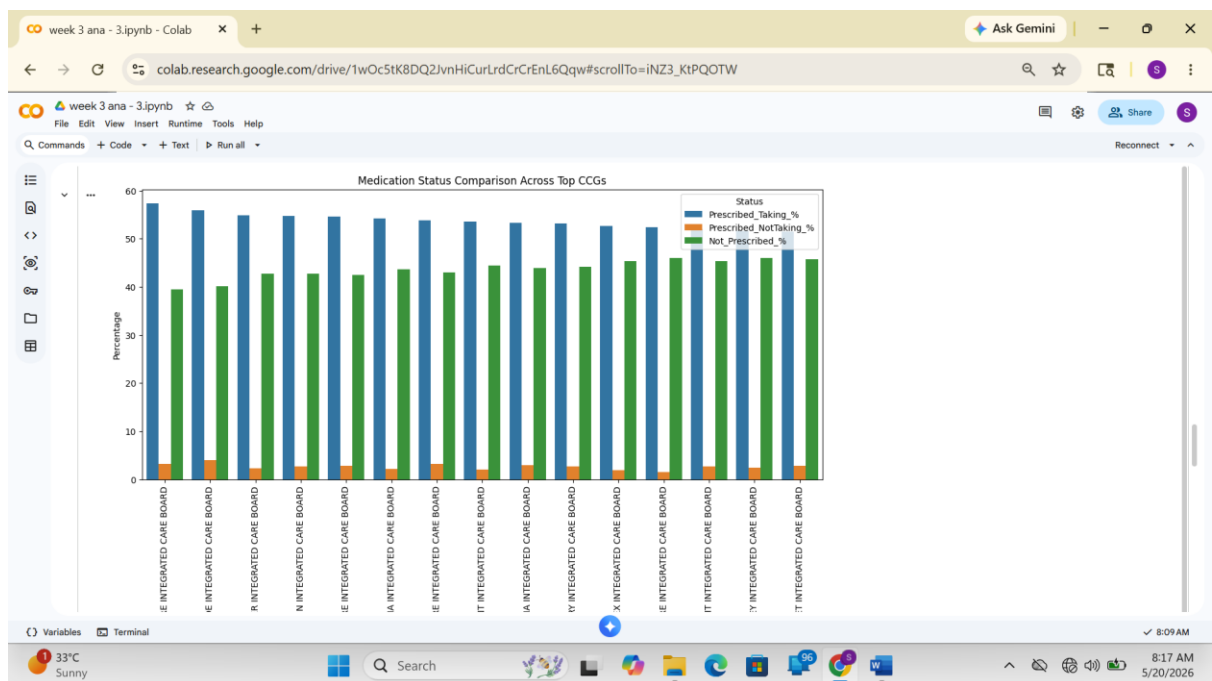
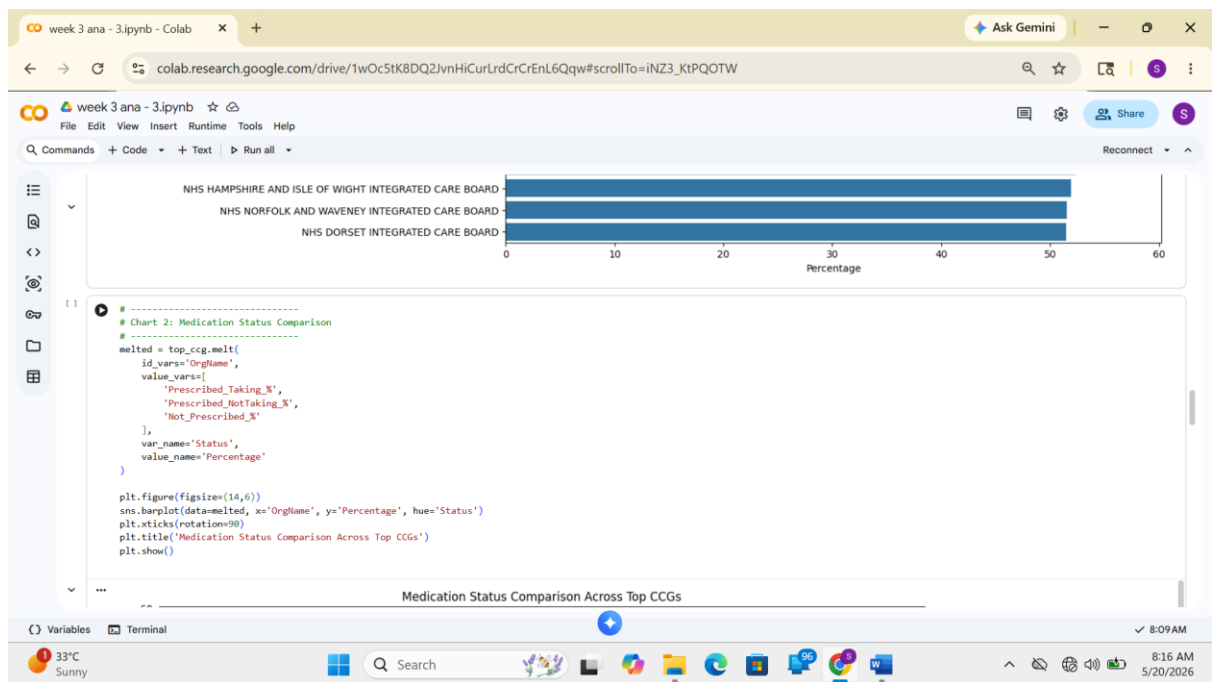
Terminal: 33°C Sunny

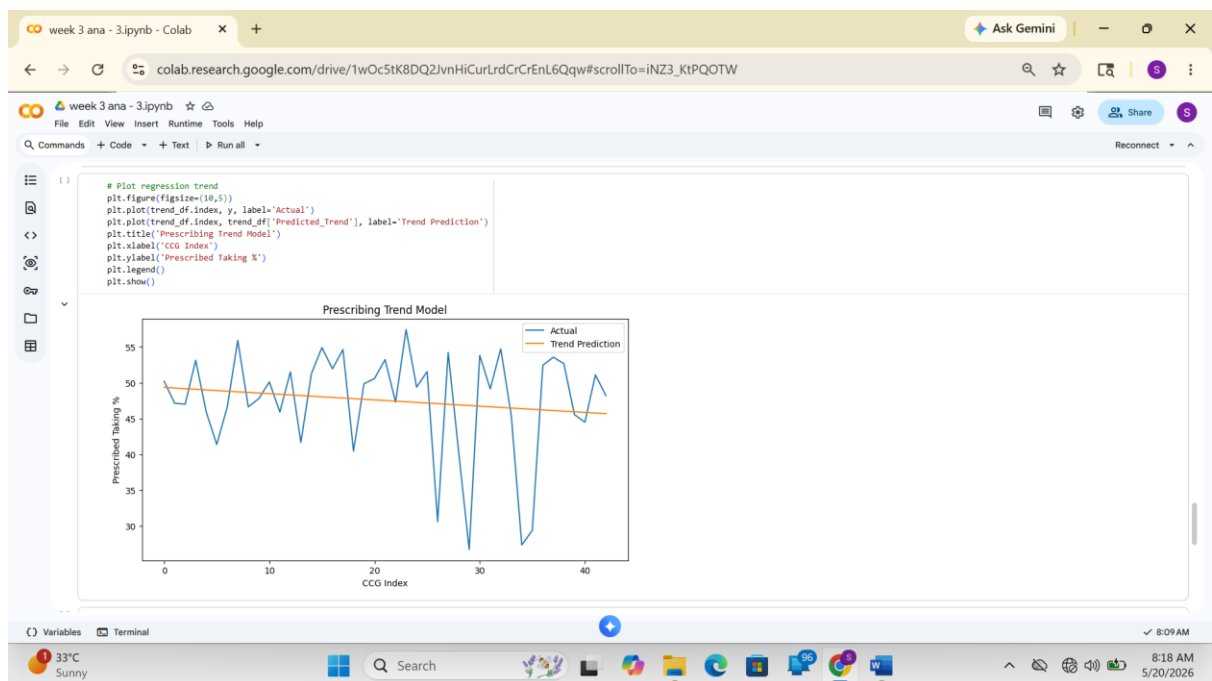
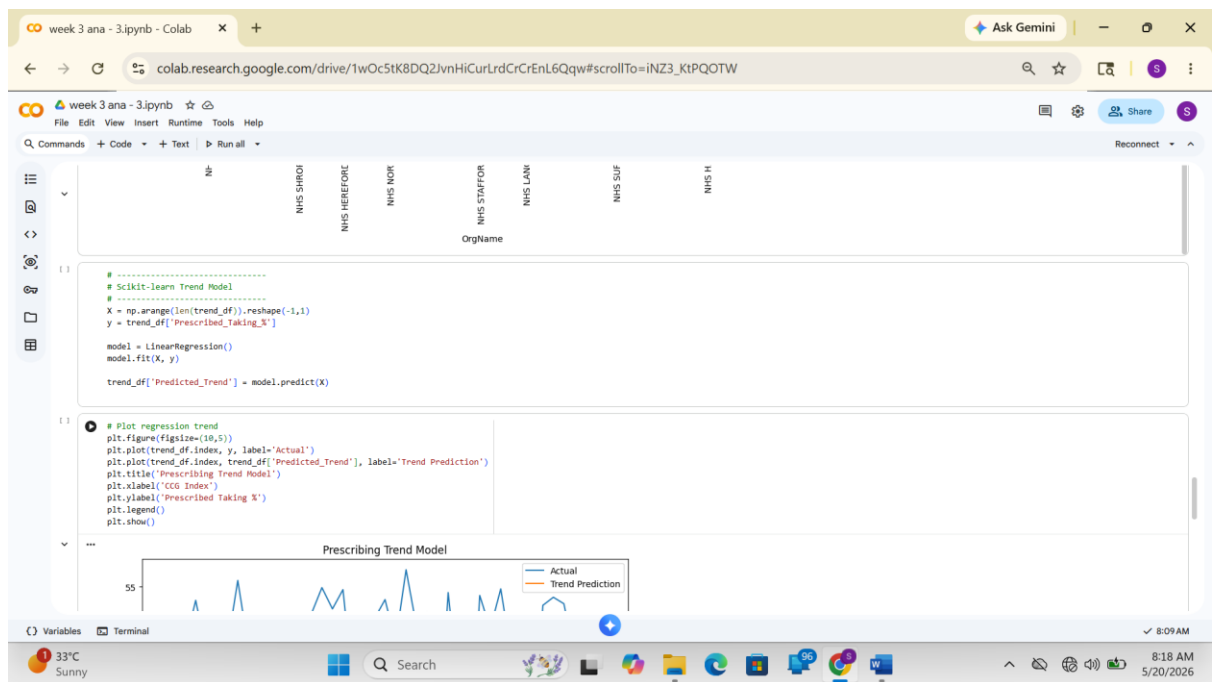
Colab interface showing code for calculating prescribing percentages and sorting:

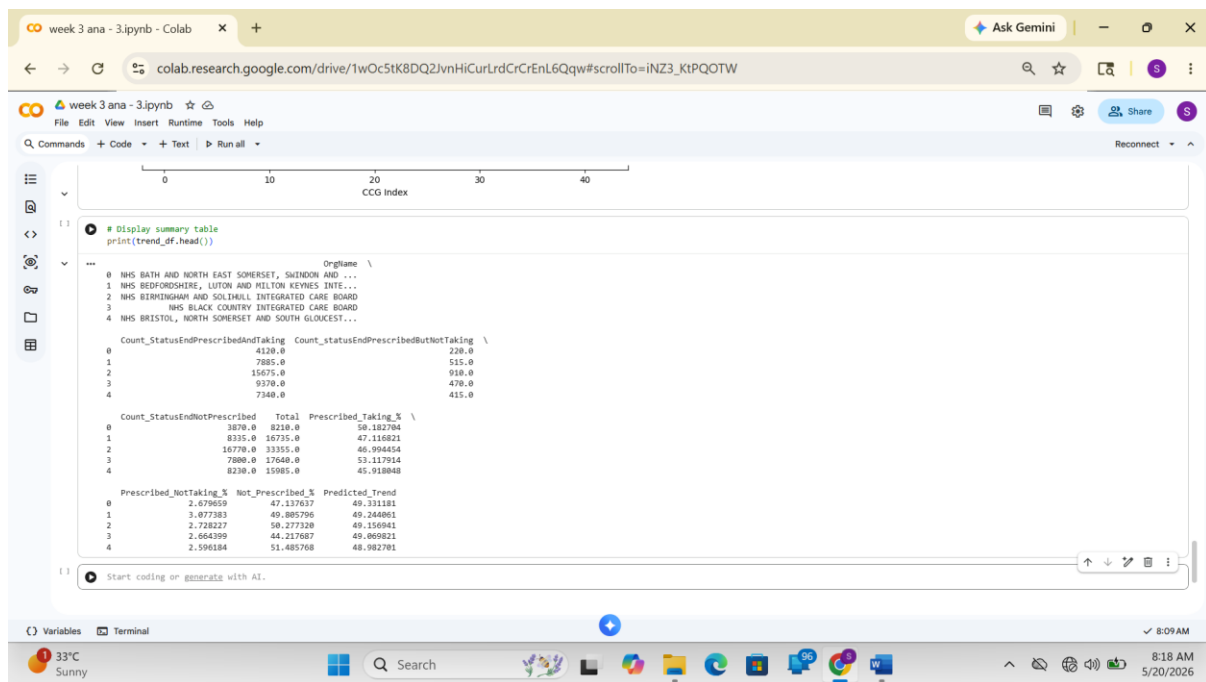
```
1 # Total patients for prescribing rate
2 trend_df['Total'] = (
3     trend_df['Count_StatusEndPrescribedAndTaking'] +
4     trend_df['Count_StatusEndPrescribedButNotTaking'] +
5     trend_df['Count_StatusEndNotPrescribed']
6 )
7
8 # Calculate prescribing percentages
9 trend_df['Prescribed_Taking_%'] = (
10     trend_df['Count_StatusEndPrescribedAndTaking'] / trend_df['Total']
11 ) * 100
12
13 trend_df['Prescribed_NotTaking_%'] = (
14     trend_df['Count_StatusEndPrescribedButNotTaking'] / trend_df['Total']
15 ) * 100
16
17 trend_df['Not_Prescribed_%'] = (
18     trend_df['Count_StatusEndNotPrescribed'] / trend_df['Total']
19 ) * 100
20
21 # Sort top 15 CCGs
22 top_ccg = trend_df.sort_values('Prescribed_Taking_%', ascending=False).head(15)
23
24 # -----
```

Terminal: 33°C Sunny









Output Charts

Chart 1: Top 15 CCGs by prescribed and taking medication percentage

Chart 2: Comparison of:

- Prescribed and taking
- Prescribed but not taking
- Not prescribed

Chart 3: Regression-based prescribing trend prediction using **Scikit-learn**

Conclusion

From this prescribing trend analysis:

- Some **CCGs/ICBs** show **higher psychotropic medication prescribing rates** than others.
- A significant proportion of patients are **prescribed and actively taking medication**, indicating ongoing pharmacological treatment.
- Certain regions show **higher non-prescribing rates**, which may indicate preference for therapy-only treatment pathways.
- The **prescribed but not taking category** helps identify adherence issues.
- Using **Scikit-learn regression**, prescribing patterns can be modelled to identify trends and outliers across organizations.

This analysis supports NHS mental health service planning by identifying **regional prescribing variation, adherence concerns, and medication usage trends**.

